Homework

Homework 2 — Approximating a root — 50 min.

Consider $u_0 = 0$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \sqrt{2 + u_n}$. Questions 1, 2 and 4 are to be done after S20; question 3 will be revisited after the lesson on continuity in S31.

1. Prove by induction that all the terms are defined and that $0 \leq u_n \leq 2$ for every $n \in \mathbb{N}$.
2. Prove by induction that $u_{n+1} \geq u_n$ for every $n \in \mathbb{N}$. Deduce that (u_n) converges.
3. Using the continuity of the function $x \mapsto \sqrt{2 + x}$, determine the limit of (u_n).
4. Prove that $0 \leq 2 - u_{n+1} \leq \frac{2 - u_n}{2}$ for every $n \in \mathbb{N}$. Deduce an upper bound for $2 - u_n$ in terms of n, then an integer N such that $0 \leq 2 - u_n \leq 10^{-3}$ for every $n \geq N$.